

Baron Eiley

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Professional Experience

Lockheed Martin Space – Sunnyvale, California

June 2024 – Present

ASIC & FPGA Hardware Design

- Developed SystemVerilog validation tests injecting simulated accelerometer sensor data into FPGA subsystems to verify signed fixed-point mathematical integration. Success criteria involved confirming velocity and distance calculations for high-altitude interceptor vehicles for USA Golden Dome portfolio.
- Validated ultra-sensitive photodetector sensor devices within hardware-in-the-loop environments to confirm critical functionality of the sensors' operation during thermal vacuum chamber testing.
- Extensive experience debugging using internal logic analyzers and oscilloscopes to confirm FPGA hardware output of complex algorithms, software defined radio FPGA hardware, and different serial communication protocols.
- Validated FPGA/SoC hardware-software integration for sensor processing algorithms on AMD Zynq Ultrascale+ platforms, verifying real-time data pipelines of image algorithms like Logarithmic Matched Filter and Complementary Median Filter implementations across PL/PS domains.
- Collaborated closely with FPGA verification engineers to solidify hardware requirements, identify test gaps, expand coverage for new and existing RTL features, and ensure designs were properly verified against customer requirements and contractual deliverables.
- Developed and executed integration and system validation workflows for FPGA/SoC platforms, validating sensor algorithm functionality, hardware-software interfaces, and end-to-end data integrity in real-time embedded environments.

Firmware Development

August 2022 – June 2024

- Developed C# UX/UI graphical user interfaces for Hardware in the loop systems to control, monitor, and validate embedded software behavior on target space rated hardware interfaced with National Instruments test equipment.
- Developed C++ code for the Ultrascale+ hard processor to interface with custom FPGA HDL designed in Vitis Model Composer for loading and unloading image data between PL and PS domains FPGAs.
Intermediate Python programming experience includes scripting for automation of FPGA HDL parsing & compile preparation for FPGA workflows.
- Developed custom scripting language using C# to command-and-control special readout integrated circuits for highly sensitive Geiger Mode Avalanche Photo-Diode sensors (GmAPD).

Education

Master of Science in Electrical Engineering

December 2025

San José State University, GPA: 3.44

San Jose, CA

Emphasis: Digital System Design and Artificial Intelligence

Bachelor of Science in Electronic Systems Engineering

May 2022

California State Polytechnic University, Pomona, GPA: 3.49

Pomona, CA

Emphasis: Digital System Design, Robotics, and Control

Graduate Projects

RapidLaneFPGA: FPGA Based Autonomous Car Highway Lane Segmentation System

- Developed a MobileNetV2 lane-segmentation CNN model and deployed it to an FPGA using AMD Vitis AI and PyTorch. The Project utilized Deep Processing Unit (DPU) hardware on 224×224 images.
- Achieved 77.10% DICE accuracy on FPGA vs. 77.8% software baseline with INT8 quantization at 10 FPS (~100ms latency) on AMD ZCU104 Ultrascale+ device, demonstrating near-real-time inference.

FPGA Based LeNET Neural Network for Digit Recognition

- Implemented LeNet-based MNIST digit classifier using Vitis High Level Synthesis and deployed FPGA.
- Hardware inference output validated against software golden data and validated FPGA inference on PYNQ-Z1, successfully classifying digits 0–9.

Swimmers Pose Estimation Using Computer Vision and AI/ML

- Developed a computer vision pipeline for swimmer pose estimation using Media-Pipe object detection, Segment Anything Model, and YOLOv7 to isolate swimmers in image and generated coordinates to train on a custom neural network.

FPGA Based Flight Controller, Search and Rescue Drone

- Developed an FPGA-based flight controller for a six-propeller search-and-rescue drone. The drone utilized a Zynq-7000 FPGA/SoC to process RC inputs and generate PWM motor control signals within the FPGA fabric.
- Designed a power distribution board for onboard camera and lighting systems.

Technical Skills

Vivado & Vitis (HLS) Verilog, SystemVerilog, & Hardware Design using C++

MATLAB/SIMULINK, Signal Processing, Control Systems Modeling, and FPGA RTL Modeling

LabView and National Instrument Equipment, Hardware Bring-up and Debug

Microsoft Excel, Word, & PowerPoint